

Who Can Turn Off Your Ethereum App?

A Full-Stack Walk-Away Test

James Carnley

Ethereum File System

When an app closes

- People keep years of work in applications.
- Pocket closed in 2025.
- Users had to recover and move their libraries.

Software people can inherit

Absence and hostile control

The team disappears

Can users continue without its services or help?

The keys change hands

Can users refuse a harmful change and continue?

The application stack

Entering the app

Names, software, identity

Continuing the work

Data, state, computation

Keeping the rules

Application control, network and settlement

Names and entry points

- ENS points to an app release.
- HTTPS gateways provide access routes.
- A retained release needs a tested independent path.

The code that actually runs

- A retained, runnable release
- Source and the inputs needed to rebuild it
- Every script the app loads after startup

Accounts and recovery

- Reading a file
- Decrypting private content
- Signing or changing shared state

Data that remains available

- Someone retains the actual bytes.
- Another operator can retrieve them.
- Private data needs decryption recovery.

File verification with EFS

- An accepted public record and expected digest
- The same file through IPFS or Arweave
- Matching bytes, altered bytes, or no response

State you can rely on

- Reachable: a provider answers
- Verified: evidence supports the claim
- Fresh: the state is recent enough
- Complete: the task has all required records

The work outside the contracts

- Required workers and external inputs
- Permission and resources to replace them
- A safe response when the work stops

Rules and change authority

Changeable rules

Who can upgrade, pause or redirect the app?

Can the user refuse?

Fixed rules

What behavior stays fixed?

How can users move if it breaks?

The network beneath the app

Connectivity and L1

Peers and node access

Inclusion and finality

Required chain history

L2 and settlement

Sequencer and proving

Available chain data

A tested recovery route

Operators and continuity

- Shared operators, cloud accounts and funding
- Retained inputs and permission to operate
- Documentation another maintainer can use

An exit someone can actually use

- A new person or a fresh machine
- Enough data, context and keys
- The original task completed elsewhere

A walk-away test for your app

- Choose one task whose loss would hurt.
- Try it without your team's services or help.
- Test what hostile authority could change.
- Fix the dependency that causes the most harm.

Their work can outlive us

Questions

Ethereum File System

Try the Sepolia app

Watch the EFS overview